

UXE Consulting AI — Output Defect Analysis

Comparison of the automated compliance report against a manual engineering review of the same project documents · Prepared as feedback for the UXE development team

Automated output under review	UXE Consulting AI, "New consultation", generated 2026-09-01T03:51:52Z. Engines: deterministic:uxe-extractive-v1 + deterministic:uxe-hashed-ngram-v1. Verdict: PARTIALLY COMPLIANT, evidence coverage 93%, confidence 45%.
Benchmark	Manual review of Chapter 10 (Smoke Control and Smoke Management Systems) against the same drawing set. 9 major findings, 6 coordination issues.
Shared inputs	Governing: uae_code_en (CDGH-OP-25, Sept 2018, 1,348 pp). Project: smoke_control_and_management_systems (Smoke Layout SM-102 to SM-111, Plot 6456327, Meydan).
Finding	4 of 5 verdicts reached are false positives caused by unit and dimension-type confusion. The 1 pass is a false negative. 55 of 60 items returned no verdict.

1. The cover metrics do not describe what they appear to describe

The report's own disclosure states that it examined the first 60 testable obligations and that a further 494 were detected but not examined. Of those 60, 55 returned "needs evidence". So the headline verdict rests on determinations for **5 requirements out of 554 — 0.9% of the code**.

Metric as displayed	What it is actually measuring	Suggested change
PARTIALLY COMPLIANT	Derived from 1 pass and 4 fails, all 5 of which are shown below to be incorrect.	Suppress the overall verdict when the determination rate falls below a threshold. Return INSUFFICIENT COVERAGE instead.
Evidence coverage 93%	Appears to measure retrieval success (a code citation was found), not requirement resolution. A reader will take it as "93% of the project is evidenced".	Split into two figures: <i>requirements resolved</i> (5/60 = 8%) and <i>corpus examined</i> (60/554 = 11%). Rename the retrieval metric.
Confidence 45%	Not tied to any user action. A 45% confidence output still emits CRITICAL directives to amend drawings.	Gate directive severity on confidence. Below ~70%, downgrade to "review" and strip the imperative wording.

2. Defect class A — numeric comparison without dimension-type checking

All four NON-COMPLIANT verdicts follow the same pattern: a number is harvested from anywhere in the drawing text, coerced to the unit named in the code clause, and compared. There is no check that the two quantities are the same *kind* of measurement, and no check that the number is semantically related to the clause.

#	Report says	Actual source of the number	Defect
11	"2.17.2.2 specifies 1.20 m but the project document states 152.40 m ." Balcony railing height. Escalated to CRITICAL .	152.4 mm = exactly 6 inches. The drawings contain an 8"x6" duct on SM-103. No 152 m dimension exists in the set.	Imperial-to-metric conversion, then mm read as m (1000× error). Value also taken from an HVAC duct and applied to a balcony guardrail clause.
15	"4.5.4.3 specifies 100 mm but the project document states 2.60 m ." Cavity fire barrier.	\A1 ; 2600 MM — the hanger rod dimension on the exhaust fan installation detail, SM-102.	Conversion correct, context wrong . A fan support dimension matched to a façade cavity barrier clause. Also note the raw CAD escape sequence was ingested.
22	"2.9.5 specifies 90.00 m but the project document states 23.73 m ." Fire lift lobby, Super Highrise threshold.	23.73 m ² is the floor area of the ground floor substation on SM-103.	Area compared against height . Dimension type is not carried through the comparison. The m ² superscript was dropped during extraction.
8	"2.15.1.3 specifies 30 min but the project document states 240 min ." Fire dampers.	Most likely a 2400 mm door height from the door schedule.	Length parsed as time. Comparison direction is also inverted : 240 exceeding a 30-minute minimum cannot be a failure even on the tool's own reading.

Root cause. The extraction layer appears to emit bare scalars without a unit class (length / area / volume / time / flow / pressure), a source context, or a link to the entity being measured. The comparator then treats any scalar as a candidate for any clause threshold. **Recommended fix:** type every extracted quantity at parse time and refuse any comparison where the dimension classes differ. This alone would have suppressed all four false positives.

3. Defect class B — the one COMPLIANT verdict is a false negative

Item 9 passes §2.15.1.5 on the strength of the annotation “WITH MOTORIZED FIRE DAMPER EX.F-BAS-04”, reasoning that “2.15.1.5 requires 120 min and the project document states 240 min, which satisfies it.”

§2.15.1.5 is the clause listing locations where fire dampers are **not** required, and Table 10.1.8.4.vi places smoke extraction ducts and stair pressurization ducts on that list. The presence of a fire damper on a smoke extract path is the **non-conformity**, not the evidence of compliance. This is the one damper annotation in the set that needed challenging, and it is the only item the tool passed.

Root cause. Clause polarity is not modelled. A “shall not be required in” clause is being read as a positive requirement, and the presence of a matching keyword is scored as satisfaction. **Recommended fix:** classify each clause as mandatory / prohibitive / exemptive at ingest, and invert the satisfaction test for the exemptive class.

4. Defect class C — absence is scored as “needs evidence” rather than as a finding

On a design review, a system that is not drawn is the finding. The tool surfaces the governing clause and then declines to conclude anything. Three examples where the clause was correctly retrieved and the conclusion was not drawn:

Item	Clause retrieved	What the tool returned	What is actually true
45	§2.6.1 Stair pressurization	NEEDS EVIDENCE — “do not evidence: control, smoke”	No stair pressurization exists anywhere in the drawing set. The riser feeds lobbies only at 700 CFM. Mandatory under Table 10.20.1.
52	§2.19 Lift hoistway pressurization	NEEDS EVIDENCE — “do not evidence: hoistway, control, general”	Not provided, and no firefighting lift lobby is drawn as an alternative. Mandatory under Table 10.20.3.
53	§2.20 Ventilation for hazardous rooms	“Inconsistent across the project documents, so a single verdict cannot be reached”	Generator room, LPG shafts, garbage and bulk waste rooms have no ventilation shown. 20 ACH required under Table 10.18.

Recommended fix. Separate two states that are currently merged. **(a) Not applicable / out of submission scope** — the clause belongs to a discipline this document does not cover. **(b) Required and absent** — the clause applies to this document type and no corresponding provision is present. State (b) is a finding and should be reported as such. Distinguishing them requires knowing the document type, which leads to defect class D.

5. Defect class D — no document-type routing

A smoke ventilation drawing set was tested against clauses on fire station bedrooms and locker rooms (item 23), above-ground and underground LPG tank installation (items 55–57), PRDP panels (58), LPG tank working pressure to ASME (59), water mist systems (32), yard hydrants (33), curtainwall perimeter joints (14), GRP façade systems (18), glass baluster design to ASCE (13) and openable window direction (12).

None of these belong in a smoke layout submission. They generate “needs evidence” rows that read to a client as gaps in the design. **Recommended fix:** classify the project document by discipline at upload, and filter the requirement set to clauses that document type can reasonably evidence. Report the remainder as out of scope rather than unevidenced.

6. Defect class E — token-overlap explanations are surfaced to end users

Rows read: “do not evidence: area, included, civil, construction, defence, horizontal” and “do not evidence: featur”. These are unmatched n-grams from the code paragraph, including a stemmed fragment. They expose the matching mechanism and give the reader nothing actionable. The recommended-actions list then repeats them verbatim as instructions — “[HIGH] Supply the section that demonstrates window, outward, see, top”.

Recommended fix: never surface raw token sets. Either express the gap in the clause's own terms, or state plainly that the requirement could not be assessed from this document.

7. Defect class F — no derived or quantitative reasoning

Every finding in the manual review that carries engineering weight required combining values from the drawing with a rule in the code. The tool performs no such derivation. Examples it could not reach:

- **Building classification.** Roof FFL +23.76 m with occupiable space → High Rise under Clause 1.7.41 → Table 10.20 governs, not Table 10.21. This single inference determines which requirement set applies to the whole submission, and nothing downstream is correct without it.
- **Air change rates.** $1,119.7 \text{ m}^2 \times \text{assumed } 3.0 \text{ m} = 3,359 \text{ m}^3$; 10 ACH → 19,770 CFM required against 18,000 CFM provided = 9.1 ACH, short.
- **Ratios.** Corridor makeup $1,400 \text{ CFM} \div \text{extract } 1,600 \text{ CFM} = 87.5\%$ against an 85% minimum.

- **Internal consistency.** On one sheet the branch ducts total 22,500 CFM, the main duct is sized for 18,000 CFM, and the grille schedule gives $20 \times 900 = 18,000$ CFM. Three irreconcilable totals for one system, all present in the same document.
- **Cross-sheet consistency.** Second, third and fourth floors all annotated +8.94 FFL, which the 3.68 m floor-to-floor spacing elsewhere in the set contradicts.

Recommended fix: a derivation layer that can (i) resolve a small set of building attributes first — occupancy, height band, sprinklered status, basement depth — and use them to select the governing tables, and (ii) evaluate arithmetic rules (area $\times$ height $\times$ ACH, ratios, sums) rather than only matching stated values. Consistency checks within and across sheets are a separate high-value class that needs no code knowledge at all.

8. What the tool did well

Three of its retrievals point at genuine items that fall outside the Chapter 10 scope of the manual review and are worth following up on this project. The retrieval layer is clearly finding relevant material — the weakness is in what happens after retrieval.

Item	Clause	Why it is worth following up
6	Ch. 5, p. 111	1-hour rated smoke barrier corridor, and smoke compartmentation for floors over 929 m ² or 465 m ² . The typical floor plate falls in that range.
39	§4.13.1 / Table 9.29	Sprinkler coverage. Determines whether smoke ductwork needs a 1-hour or 2-hour rating.
40	§4.14	Typical-floor electrical rooms under 10 m ² require 1-hour rated enclosures. The floor plans show 5.12 m ² electrical rooms.

Citation quality is also good: chapter, section, page and verbatim quotation, with a pinned source version. That traceability is the right foundation — the defects above sit in the reasoning layer built on top of it.

9. Suggested priority for the development team

Pri.	Change	Defect class addressed	Effect on this report
1	Type every extracted quantity by dimension class and block cross-class comparisons.	A	Removes all 4 false positives.
2	Gate directive severity on confidence; suppress the overall verdict below a determination-rate threshold.	—	Prevents a 45%-confidence run from issuing CRITICAL amend-the-drawing instructions.
3	Model clause polarity (mandatory / prohibitive / exemptive).	B	Converts the single false pass into a correct finding.
4	Route by document discipline; separate “out of scope” from “required and absent”.	C, D	Removes roughly 20 spurious rows; converts 3 non-verdicts into real findings.
5	Replace token-overlap explanations with clause-language gaps.	E	Removes unusable directives from the recommended actions list.
6	Add a derivation layer: building attribute resolution, arithmetic rules, internal consistency checks.	F	Reaches the classification, ACH and reconciliation findings — the highest-value class, currently absent.

Summary. The retrieval and citation layers are sound. The reasoning layer is emitting high-severity, high-confidence-sounding directives from untyped numeric matches. On this run, following the four CRITICAL and HIGH “amend the project document” actions would have produced entirely wasted rework — including amending balcony railings on the basis of a duct dimension — while the nine real non-conformities in the same drawing set went unreported. Priorities 1 and 2 are the highest-leverage changes and are independent of any code-domain work.

Prepared from the UXE Consulting AI output dated 2026-09-01 and a manual review of the same drawing set against CDGH-OP-25 (September 2018), Chapter 10. Clause interpretations reflect that edition; the project’s building permit is dated 12.2024, so the governing edition should be confirmed independently. This document is intended as engineering feedback on tool output and is not a compliance certificate for the project.